Senanayak Sesh Kumar Karri

Research name: K. S. Sesh Kumar

Machine Learning Researcher | Research Fellow, Imperial College London

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Research Summary

Machine-learning researcher at Imperial College London specialising in probabilistic modelling, graph-structured learning, geometry-aware neural networks, and convex and submodular optimisation. PhD in machine learning from INRIA / École Normale Supérieure (SIERRA), followed by research appointments in statistical machine learning and data science. Research published at NeurIPS, ICML, JMLR, TMLR, and *Machine Learning*. Finance is an application domain for my methodological work: I study learning under non-stationarity, structured high-dimensional estimation, and reinforcement learning for sequential decisions, with applications to financial time series, covariance estimation, and portfolio management.

Accepted Papers at NeurIPS 2026

[1] **K. S. Sesh Kumar** and V. Borovitskiy, **Practical Non-Stationary Graph Gaussian Processes**. *Fortieth Conference on Neural Information Processing Systems (NeurIPS 2026), accepted as a poster*, submission 25512.

[2] **K. S. Sesh Kumar**, **Geometry-Constrained Kolmogorov–Arnold Networks: Learning Edge Geometry via Banach Duality**. *Fortieth Conference on Neural Information Processing Systems (NeurIPS 2026), accepted as a poster*, arXiv:2608.25807.

Academic Positions

Research Fellow (Machine Learning Research) — London, UK
Brevan Howard Centre for Financial Analysis, Imperial College Business School — *Sept 2022 – present*

Research Fellow, Data Science Institute — London, UK
Imperial College London — *Sept 2019 – Aug 2022*

Research Fellow, Statistical Machine Learning Group — London, UK
Imperial College London; Leverhulme Centre for the Future of Intelligence fellowship — *Dec 2017 – Aug 2019*

Postdoctoral Researcher — Klosterneuburg, Austria
Institute of Science and Technology (IST) Austria — *Oct 2016 – Oct 2017*

Education

PhD in Machine Learning, SIERRA, INRIA / École Normale Supérieure — Paris, France
May 2013 – Sept 2016

- Thesis: *On the Links between Probabilistic Graphical Models and Submodular Optimisation*, École Normale Supérieure, 2016.

MSc, Mathematics, Vision, Learning (MVA), ENS Cachan — Cachan, France
Sept 2012 – Apr 2013

B.Tech., Computer Science and Engineering, IIIT Hyderabad — Hyderabad, India
Jul 1999 – Aug 2003

Selected Publications

[1] M. Conserva, M. P. Deisenroth, and **K. S. Sesh Kumar**, **The Graph Cut Kernel for Ranked Data**. *Transactions on Machine Learning Research (TMLR)*, 2022.

[2] C. Buizza, C. Quilodrán Casas, P. Nadler, J. Mack, S. Marrone, Z. Titus, C. Le Cornec, E. Heylen, T. Dur, L. Baca Ruiz, C. Heaney, J. Amador Díaz López, **K. S. Sesh Kumar**, and R. Arcucci, **Data Learning: Integrating Data Assimilation and Machine Learning**. *Journal of Computational Science*, 2021.

[3] R. Moriconi, M. P. Deisenroth, and **K. S. Sesh Kumar**, **High-Dimensional Bayesian Optimization with Manifold Gaussian Processes**. *Machine Learning*, 2020.

[4] **K. S. Sesh Kumar**, F. Bach, and T. Pock, **Fast Decomposable Submodular Function Minimization using Constrained Total Variation**. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[5] R. Moriconi, **K. S. Sesh Kumar**, and M. P. Deisenroth, **High-Dimensional Bayesian Optimization with Projections using Quantile Gaussian Processes**. *Optimization Letters*, 2019.

[6] **K. S. Sesh Kumar** and M. P. Deisenroth, **Differentially Private Empirical Risk Minimization with Sparsity-Inducing Norms**. *Privacy Preserving Machine Learning (PPML) Workshop, NeurIPS*, 2018.

[7] **K. S. Sesh Kumar** and F. Bach, **Active-Set Methods for Submodular Minimization Problems**. *Journal of Machine Learning Research (JMLR)*, 2017.

[8] **K. S. Sesh Kumar** and F. Bach, **Maximizing Submodular Functions using Probabilistic Graphical Models**. *Discrete Optimization for Machine Learning (DISCML) Workshop, NIPS*, 2013.

[9] **K. S. Sesh Kumar** and F. Bach, **Convex Relaxations for Learning Bounded Treewidth Decomposable Graphs**. *International Conference on Machine Learning (ICML)*, 2013.

Preprints and Working Papers

[1] F. Allen, M. T. Kacperczyk, and **K. S. Sesh Kumar**, **Adaptive Market Ecology and Conditional Strategy Performance: Evidence from Machine-Learning Market-Timing Strategies**. *SSRN working paper*, 2026.

[2] F. Allen, M. T. Kacperczyk, and **K. S. Sesh Kumar**, **Modern Machine Learning Tools in Finance: A Critical Perspective**. *SSRN working paper*, 2025.

[3] S. Cohen, **K. S. Sesh Kumar**, and M. P. Deisenroth, **Sliced Multi-Marginal Monge Optimal Transport**. *arXiv:2102.07115*, 2021.

[4] **K. S. Sesh Kumar**, A. Barbero, S. Jegelka, S. Sra, and F. Bach, **Convex Optimization for Parallel Energy Minimization**. *arXiv:1503.01563 / HAL 01123492*, 2015.

Awards and Fellowships

- 2019 — Research Fellowship, Data Science Institute, Imperial College London.
- 2018 — Research Fellowship, Leverhulme Centre for the Future of Intelligence.
- 2017 — Data Science Institute research funding, Imperial College London.

Earlier Research and Industry Experience

Research Engineer, WILLOW team, INRIA, Paris (2012–2013); Research Intern, SIERRA team, INRIA, Paris (2012); Research Intern, Computer Vision and Geometry Group, ETH Zurich (2011); Research Engineer, Télécom SudParis, Evry (2010–2011); GPGPU Engineer, AMD Research and Development, Hyderabad (2006–2010); Research Assistant, CVIT, IIIT Hyderabad (2003–2006), with publications at ICDAR, ICVGIP, DAS, and PReMI.

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14 October 2026

Application for the G-Research NeurIPS 2026 travel and conference grant

Dear Travel Grant Committee,

I am applying for support to attend NeurIPS 2026 in Sydney, where I have two papers accepted for poster presentation. I am a machine-learning researcher and Research Fellow at Imperial College London, based at the Brevan Howard Centre for Financial Analysis, Imperial College Business School. My research centres on probabilistic modelling, learning on graphs, neural-network geometry, and optimisation; financial markets provide an application domain for these methods.

My accepted papers contribute to core machine-learning methodology:

- **Practical Non-Stationary Graph Gaussian Processes** (first author; submission 25512). This work develops Gaussian-process models for heterogeneous graphs, linking node-level variability to interaction strength through an interpretable parameter. It addresses how probabilistic models can represent variation across a network rather than impose a uniform structure.
- **Geometry-Constrained Kolmogorov–Arnold Networks: Learning Edge Geometry via Banach Duality** (sole author; arXiv:2608.25807). This work derives learnable edge functions from Banach-space duality, allowing the geometry of each edge to adapt to data. It investigates interpretable representations and learning from noisy or limited observations.

I completed my PhD in machine learning at INRIA / École Normale Supérieure, working on probabilistic graphical models and submodular optimisation. Subsequent research appointments at IST Austria and Imperial's statistical machine-learning group and Data Science Institute developed this methodological focus. My publications include work at NeurIPS, ICML, JMLR, TMLR, and *Machine Learning*. My current financial applications concern non-stationary time series, structured covariance estimation, and reinforcement learning for portfolio decisions.

Grant support would help meet the cost of travelling from London to Sydney and participating in the conference. Attendance would allow me to present both papers, obtain feedback from researchers in probabilistic modelling and neural architectures, and develop collaborations around these research questions.

Thank you for considering my application. My curriculum vitae accompanies this letter.

Yours sincerely,
Senanayak Sesh Kumar Karri